

Healthy Skepticism · Session 2 take-home summary

Fault 2: probability reversal

Two probabilities can sound like the same sentence and be different numbers. One of them is what a test, a study or an observation can actually measure. The other is what you wanted to know when you asked.

What can be measured. $P(\text{observation} \mid \text{condition})$. When the condition is there, how often do we see this?

What you want to know. $P(\text{condition} \mid \text{observation})$. Having seen it, how likely is the condition?

The fault is reading one as if it were the other. How far apart the two sit depends on something the observation never tells you, which is how common the condition is in the first place.

Counting with a waffle chart

Everything in this session is done by counting. A waffle chart is a hundred squares, and each square is one member of the population. Give the population two attributes, call them A and B, and every square falls into one of four groups.

In the example we used, 18 squares have both A and B, 2 have A only, 12 have B only, and 68 have neither. The four groups add to 100, because every square is counted once. Written as probabilities, $P(A \text{ and } B)$ is 18%, $P(A \text{ only})$ is 2%, $P(B \text{ only})$ is 12%, and $P(\text{neither})$ is 68%. A probability, in this course, is just a name for a share of the grid.

The percentages here are illustrative, chosen so the arithmetic stays visible.

What the bar means

The vertical bar in $P(A \mid B)$ is read as given, and it's an instruction about who stays in the picture. It tells you to stop looking at the whole grid and look at part of it.

Ask for $P(B \mid A)$ and you throw away the 80 squares that aren't A. Twenty are left, and 18 of them are B, so the answer is 90%. Ask for $P(A \mid B)$ and you throw away the 70 squares that aren't B. Thirty are left, and 18 of them are A, so the answer is 60%.

The same 18 squares sit in the numerator both times. What changed is the group you divided by. That's the whole of the reversal, and it's worth holding onto, because every example that follows is this picture with names attached.

The wet lawn

Take A to be rain overnight and B to be a wet lawn in the morning, and the grid becomes a record of 100 mornings. It rained on 20 of them. The lawn was wet on 30.

$$P(\text{lawn wet} \mid \text{rain}) = 18 / 20 = 90\%$$

$$P(\text{rain} \mid \text{lawn wet}) = 18 / 30 = 60\%$$

Rain almost always wets the lawn. A wet lawn is a good deal less telling, because sprinklers and dew wet it too. Nothing about the mornings changed between those two numbers. Only the group being counted changed.

From the lawn to the clinic

Every diagnostic test reports two numbers, and each one is a forward probability that gets flipped in the room.

Sensitivity. $P(\text{test positive} \mid \text{have the cause})$, which is what the test was measured on. It gets read as $P(\text{have the cause} \mid \text{test positive})$.

Specificity. $P(\text{test negative} \mid \text{no cause})$, also measured. It gets read as $P(\text{no cause} \mid \text{test negative})$.

Each reversal is undone by the other kind of mistake. A positive result is undone by the false positives, and a negative result is undone by the false negatives. There's an old mnemonic for the direction each one is safe in: a sensitive test with a negative result rules out, and a specific test with a positive result rules in.

Both reversals live in every test. We took each one in the place where it does the most damage. A screening mammogram is good at finding the cancers that are there, so it's the positive result that misleads. A COVID rapid test is good at clearing people who are well, so it's the negative that misleads.

A screening mammogram: if it's positive, do I have cancer?

What the test is measured on

Sensitivity is 87%. Of every 100 women who have cancer, 87 are called back and the other 13 are missed. That's a good number, and it's the number the mammogram is judged on. Notice where it starts, though. It starts from already knowing she has cancer, which is the only direction a test can be measured in.

A positive mammogram is a callback. The radiologist wants more pictures. Nothing has been found yet.

Sensitivity 86.9%. Lehman CD et al. National performance benchmarks for modern screening digital mammography, Breast Cancer Surveillance Consortium. Radiology 2017;283(1):49-58. 1,682,504 screens, 2007-2013.

Ten thousand women, screened once

Six women in a thousand have cancer, so 59 of the 10,000 have it, and that figure is an average across ages 40 to 89. It climbs with age, nearer 4 per 1,000 in the fifties and 6 or 7 in the sixties. Applying the benchmark rates gives four groups:

- has cancer, called back: 51
- has cancer, not called back: 8
- no cancer, called back anyway: 1,103
- no cancer, all clear: 8,838

On a waffle chart the band of women called back who don't have cancer is eleven rows deep. It's more than twenty times the size of the group the test is looking for, and that band is what makes the reversal so large here.

Cohort of 10,000 is illustrative; the rates applied to it are the published benchmarks. Lehman CD et al., Radiology 2017;283(1):49-58.

The two questions, on the same picture

Look only at the 59 women who have cancer and set everyone else aside. Of those 59, the ones called back number 51.

$$P(\text{callback} \mid \text{cancer}) = 51 / 59 = 87\%$$

Now look only at the women who were called back. There are 1,154 of them, and 51 have cancer.

$$P(\text{cancer} \mid \text{callback}) = 51 / 1,154 = 4.4\%$$

Same picture, same 51 women in the numerator. We divided by a different group, and twenty-two of every twenty-three callbacks turn out to be false alarms. The 4.4% isn't our arithmetic, incidentally. It's the positive predictive value the benchmark study itself publishes.

Positive predictive value of an abnormal interpretation 4.4%; of a biopsy recommendation 25.6%. Lehman CD et al., Radiology 2017;283(1):49-58.

What the letter means

87% is what the mammogram was measured on. 4% is what her callback letter means. Same test, same day, same woman. Unlikely is why she shouldn't panic. Not ignorable is why the biopsy is warranted.

Why reasonable people disagree

Of US women aged 50 who are screened every year for ten years, somewhere between 0.03% and 0.32% avoid a breast cancer death, between 49% and 67% have at least one false alarm, and between 0.3% and 1.4% are overdiagnosed and treated for a cancer that would never have troubled them.

Nobody in this argument disputes the sensitivity, the specificity or the 4.4%. The disagreement is over how to weigh a false alarm for half of them against a death averted for a fraction of one percent. So the guidance splits. The American College of Radiology recommends annual screening from 40. The Task Force says every two years from 40. The Nordic Cochrane group and the Swiss Medical Board have argued for not running screening programmes at all.

Ranges from published estimates for women aged 50 screened annually for a decade. Overdiagnosis is contested: the Canadian National Breast Screening Study's 25-year follow-up put 106 of 484 screen-detected cancers, or 21.9%, as overdiagnoses; Gotzsche and Jorgensen, Cochrane 2013 (CD001877), found no effect on overall mortality across ten trials. An 11.6% callback rate compounded over ten independent screens gives about 71%, near the top of the false-alarm range.

A COVID rapid test: if it's negative, am I clear?

What the swab is measured on

Specificity is 99.1%. Of every 100 people who are well, 99 test negative, so almost nobody healthy is wrongly flagged. That's about as specific as any test in common use, and it's why a positive swab is

worth believing. But look where that number starts. It starts from already knowing the person is well, and nobody holding a negative result knows that.

Rapid antigen tests in symptomatic people: specificity 99.1%, sensitivity 73.0% (95% CI 69.3 to 76.4). Dinnes et al., Cochrane Database of Systematic Reviews 2022;7:CD013705.

Ten thousand people with symptoms

Everyone in this group feels ill during a wave, so the odds before testing are far higher than in the street outside. Take four in ten to have COVID:

- has COVID, tests positive: 2,920
- has COVID, tests negative: 1,080
- no COVID, tests positive: 54
- no COVID, tests negative: 5,946

The band to watch is the 1,080. Eleven rows of people who had COVID and were told they didn't.

The pre-test probability of 4 in 10 is illustrative, chosen to represent symptomatic testing during a wave. Test figures from Dinnes et al., Cochrane 2022;7:CD013705.

What a negative means

There are 7,026 negatives in that cohort, and 1,080 of them have COVID.

$$P(\text{negative} \mid \text{no COVID}) = 99\%$$

$$P(\text{no COVID} \mid \text{negative}) = 85\%$$

99% is what the swab was measured on. 85% is what your negative means, which is nowhere near certainty. More than one person in seven who was told they were negative had COVID and went out anyway. A negative narrows the odds. It doesn't clear you.

When nobody catches it

A rape trial in England, conviction quashed in 1994

The forensic expert told the court that the likelihood of the sample coming from any man other than the accused was one in three million. That number describes how rare the profile is. It was offered as the chance he was innocent, the judge repeated it to the jury, and the Court of Appeal quashed the conviction.

R v Deen, Court of Appeal, reported in The Times, 10 January 1994. Rules on presenting DNA statistics were later set in Doheny and Adams [1997] 1 Cr App R 369.

Traffic stops in Houston, 2004 to 2015

A two-dollar roadside kit turns blue for cocaine. It also turns blue for dozens of other things, which has been known since the 1970s. Blue was taken to mean cocaine. The lab checked only afterwards, and more than 250 people had already pleaded guilty to possessing nothing.

Houston Forensic Science Center and the Harris County conviction integrity unit, reported by ProPublica and The New York Times Magazine, 2016.

Both times, a number about how often this happens to the innocent was read as the chance of innocence.

The detector that cannot tell

Turnitin has claimed 98% accuracy for its AI writing detector. Copyleaks 99.1%, GPTZero 99%, Winston AI 99.98%. Every one of those is a statement about machines: if this text came from an AI, we'll flag it. What a flagged student needs is the other direction. Of the essays you flag, how many were actually written by an AI?

Vanderbilt ran that number. Even at a 1% false positive rate, their 75,000 submissions a year would falsely accuse 750 students, and they switched the tool off in August 2023. A 1% rate turns out to be generous. Stanford researchers ran seven detectors over 91 TOEFL essays, every one of them written by a human, and 61.3% were flagged as machine-written, with about one in five flagged by all seven. On essays by native English speakers the same detectors almost never erred.

It's the same flip, with a transcript at the end of it. A number about how often the tool catches machines, read as the chance that a student cheated.

Liang W, Yuksekgonul M, Mao Y, Wu E, Zou J. GPT detectors are biased against non-native English writers. Patterns 2023. Vanderbilt disabled Turnitin's AI detection in August 2023. Vendor accuracy figures are the vendors' own marketing claims.

What to ask

The trap. What the test reports gets flipped into what you asked. A positive is read as proof, and a negative as the all-clear.

The question. Out of everyone who got this result, how many actually have it?

That question can't be answered from the test alone. It needs one more thing, which is how common the condition is among people like you. Ask for it by name: the prevalence, or the pre-test probability. If nobody can tell you, the reversal can't be computed, and any confident answer you're given is an assumption wearing a number.

Next session

Arbitrary categories. A line drawn across a continuous measurement gets read as a fact about your body. The question to carry in is where you sit on the curve, rather than which side of the line.

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